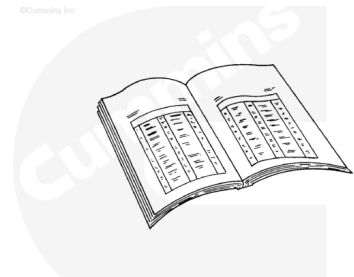


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

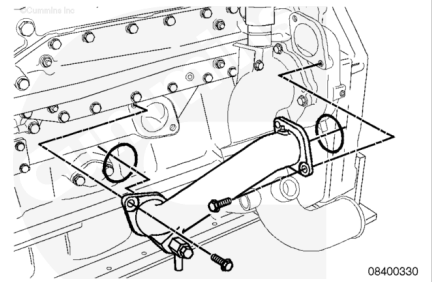
- Disconnect the batteries or air supply line to the air starter. Refer to Procedure 013-007 in Section 13. (</qs3/pubsys2/xml/en/procedures/89/89-013-007.html>)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html>)

Remove

Remove the four water pump outlet pipe capscrews.

Remove the water pump outlet pipe.

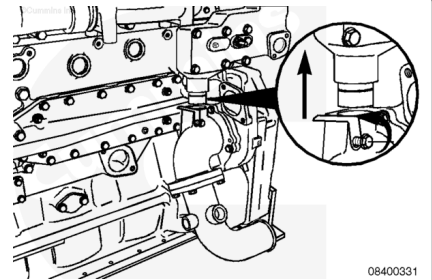
Discard the o-rings.



Loosen the bypass tube clamp.

Slide the bypass tube up into the thermostat housing.

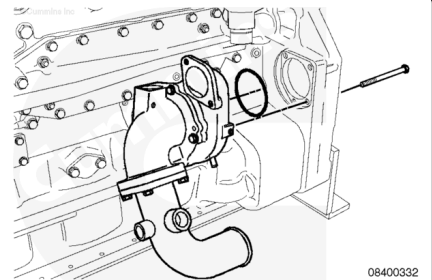
Note : Some components shown removed for clarity.



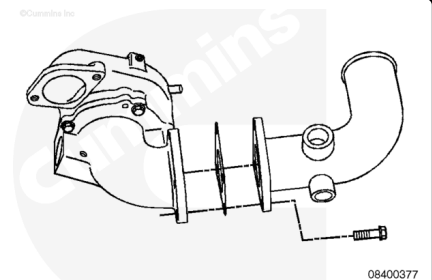
Remove the four water pump capscrews.

Remove the water pump.

Discard the o-ring.



Remove the water pump inlet connection and discard the gasket.

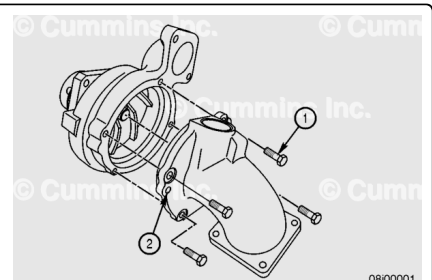


Disassemble

Remove the four capscrews (1) securing the inlet housing to the pump.

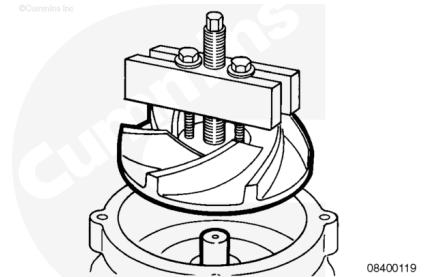
Remove the inlet housing. If necessary, use two capscrews (2) in the threaded holes (2) to force the assembly apart.

Remove and discard the large and small o-rings.



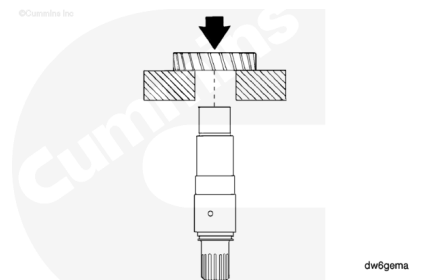
Note : The jack screw in the puller must pass easily through the impeller bore to prevent damage to the impeller.

Use a standard puller, Part Number ST-647 or equivalent, to remove the impeller.

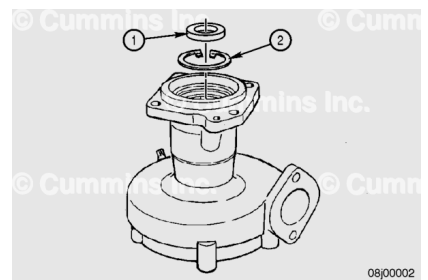


Use a water pump bearing separator, Part Number 3375326 or equivalent, to support the gear.

Use an arbor press. Press the shaft and water pump assembly out of the gear.

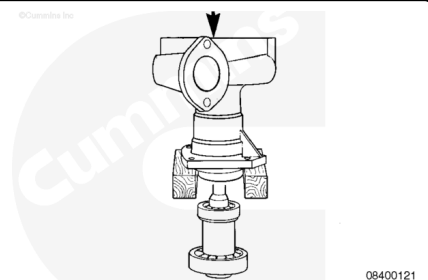


Remove the thrust ring spacer (1) and the large retaining ring (2).

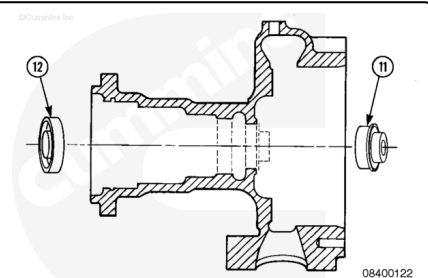


Support the water pump body, as illustrated.

Use an arbor press to push on the impeller end of the shaft. Remove the bearing and shaft assembly.



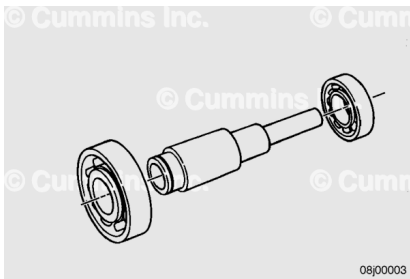
Remove and discard the water seal (11) and oil seal (12).



Before removing the shaft from the bearings, check the bearings.

If necessary, press the shaft from the bearings.

Discard the bearings.



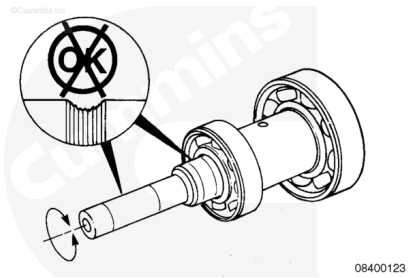
Clean and Inspect for Reuse

Inspect the bearing and shaft assembly for wear in the seal areas.

Spin the bearings to check for roughness.

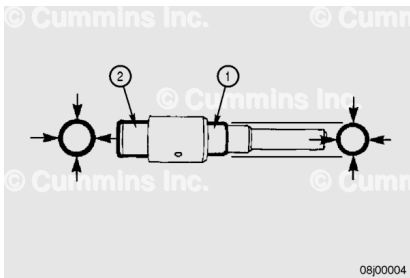
If the grooves can be felt with a fingernail, the shaft **must** be replaced.

If damage to the shaft or bearings are found, they **must** be replaced.



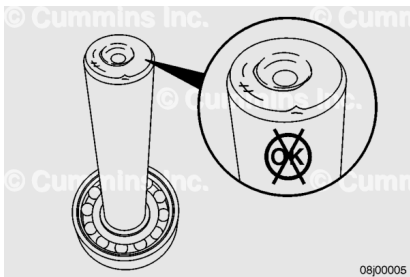
If the bearings were removed, measure the shaft diameter at the bearing journal locations.

Shaft Diameter (Callout 1 and 2)		
mm		in
25.002	MIN	0.9843
25.011	MAX	0.9847



Clean and inspect the end of the shaft.

Remove any burrs or debris caused by removal of the impeller, gear, or bearings.

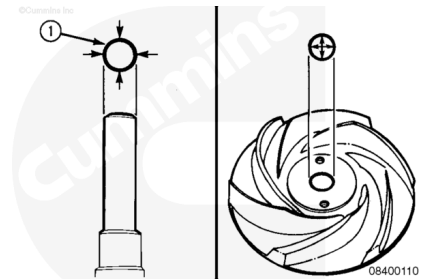


Use the following formula to determine the press fit.

Press fit = shaft outside diameter (1) minus impeller inside diameter.

Impeller to Shaft Press Fit

mm		in
0.03	MIN	0.001
0.07	MAX	0.003

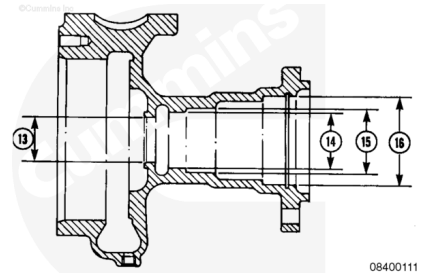


Inspect the impeller press fit area of the shaft for nicks or scratches. If any marks can be felt with a finger nail, the shaft **must** be replaced.

Measure the inside diameter of the water pump housing bore.

Housing Bore Diameter

	mm		in
(13) Water Seal	46.00	MIN	1.811
	46.25	MAX	1.821
(14) Oil Seal	42.00	MIN	1.654
	42.09	MAX	1.657
(15) Rear Bearing	51.985	MIN	2.0467
	52.004	MAX	2.0474
(16) Front Bearing	61.967	MIN	2.4396
	61.986	MAX	2.4404



Assemble

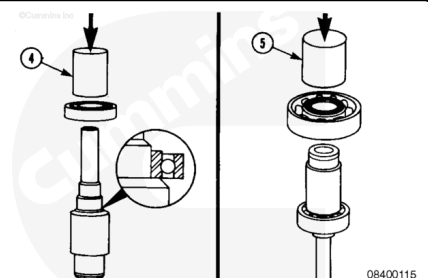
⚠ CAUTION ⚠

The mandrel must make contact on the inner race of the bearing to prevent damage to the bearing.

If removed, install the new bearings onto the shaft.

Support the shaft.

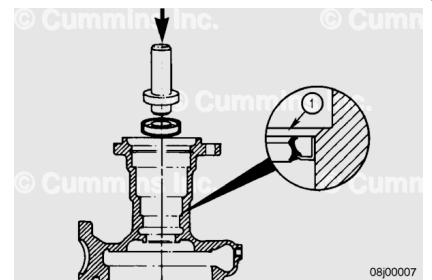
Use an arbor press and a water pump bearing mandrel to install the bearings.



Note : The smaller bearing (4) is pressed on the journal, closer to the middle of the shaft, while the bigger bearing (5) is closer to the end of the shaft.

Use the water pump seal driver to install the new oil seal with the stepped lip facing down toward the impeller.

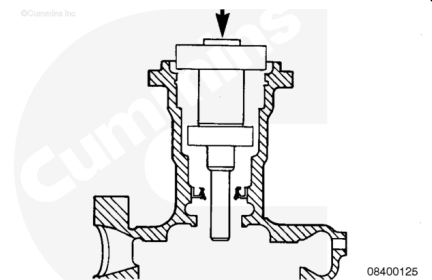
The seal must be installed so that the top of the seal is flush with the bottom edge of the chamfer (1) on the water pump body.



Support the water pump housing.

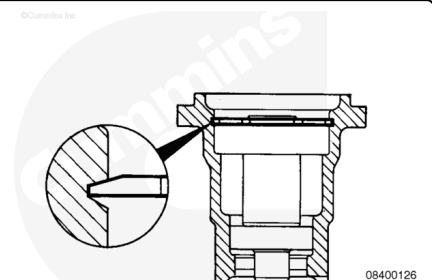
Use an arbor press to push on the end of the shaft. Install the shaft and bearing assembly.

After installing the shaft, flip the assembly over and check that the oil seal has **not** moved.



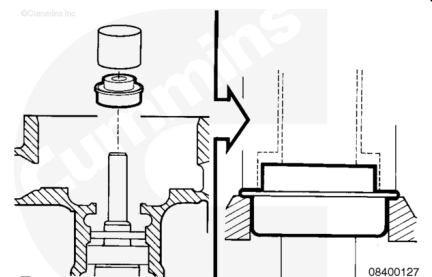
The beveled edge **must** be positioned as illustrated.

Install the large retaining ring.



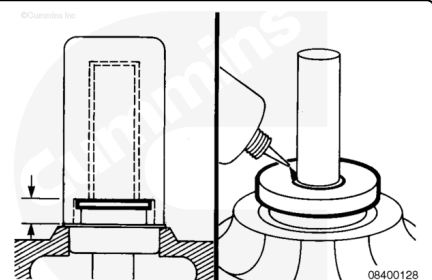
Use a water pump seal driver to install the water seal.

The shoulder on the seal **must** touch the housing.



⚠ CAUTION ⚠

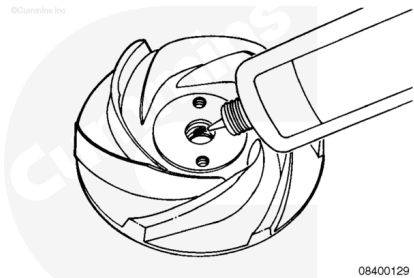
More than one drop of Loctite® 290, or equivalent, on the seal seat will glue the faces of the seal together, resulting in damage to the seal.



Use an arbor press and water pump seal driver to install the seal seat.

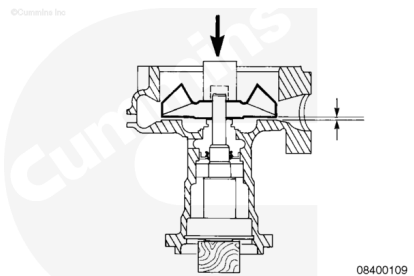
Apply one drop of Loctite® 290, or equivalent, to the seal seat, as illustrated.

Apply a smooth coating of Loctite® 290, or equivalent, to the inside diameter of the impeller, as illustrated.



Use an arbor press and a mandrel to install the impeller to the specified clearance.

Use a feeler gauge in the water outlet port to measure the clearance.



Impeller Vane-to-Body Clearance

mm		in
2.50	MIN	0.098
2.90	MAX	0.114

Note : Spin the shaft and check the clearance in multiple locations to make sure the water pump has been assembled correctly.

⚠ CAUTION ⚠

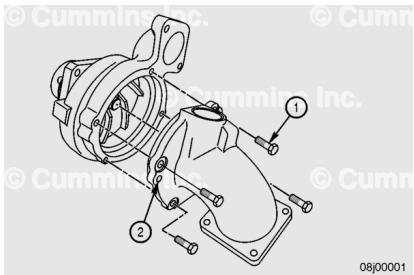
Do not use engine oil on the o-rings as it may cause them to swell.

Use vegetable oil to lubricate the o-rings (2).

Install the parts as illustrated.

Install the capscrews (1).

Torque Value: 65 n•m [48 ft-lb]

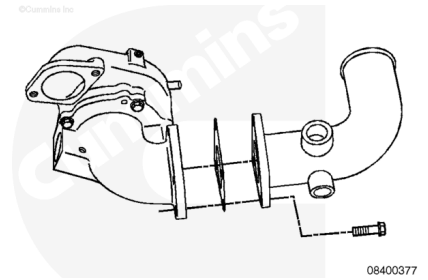


Install

Install the inlet connection, new gasket, and capscrews.

Tighten the capscrews.

Torque Value: 70 n•m [50 ft-lb]

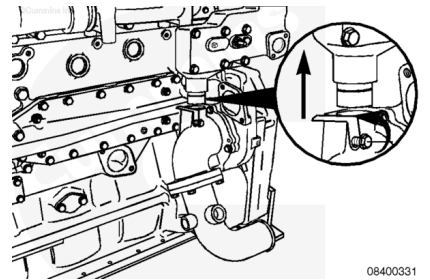


Install new o-rings on the bypass tube.

Lubricate the o-rings with vegetable oil.

Slide the tube up into the thermostat housing.

Note : Some components are shown removed for clarity.



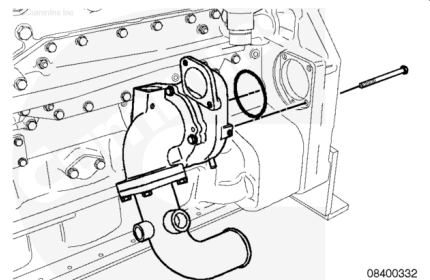
Install a new o-ring on the water pump.

Lubricate the o-ring with clean engine oil.

Install the water pump and capscrews.

Tighten the capscrews.

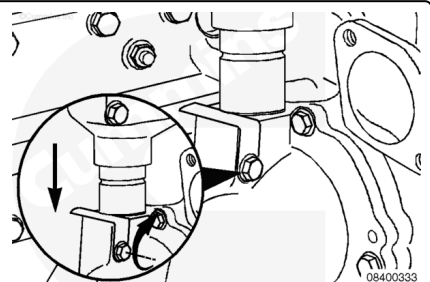
Torque Value: 113 n•m [83 ft-lb]



Slide the bypass tube down from the thermostat housing into the water pump.

Slide the bypass tube clamp into the groove in the bypass tube and tighten the capscrew.

Torque Value: 66 n•m [49 ft-lb]



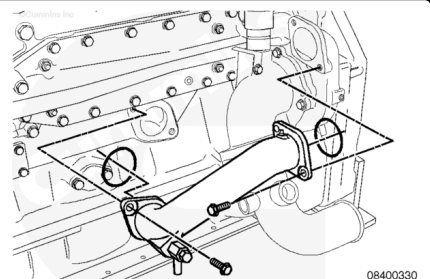
Install new o-rings on the water pump outlet pipe.

Lubricate the o-rings with vegetable oil.

Install the water pump outlet pipe and capscrews.

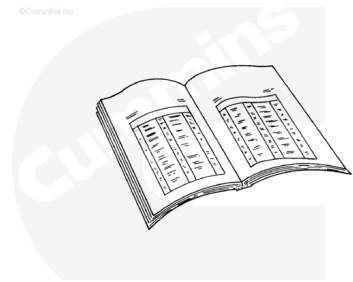
Tighten the capscrews.

Torque Value: 113 n•m [83 ft-lb]



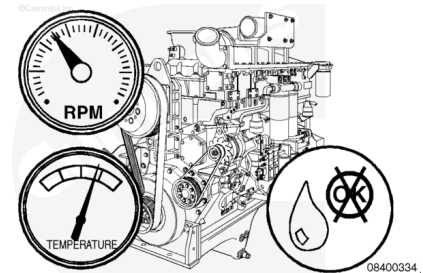
Finishing Steps

- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html>)
- Connect the batteries or air supply line to the starter. Refer to Procedure 013-007 in Section 13. (</qs3/pubsys2/xml/en/procedures/89/89-013-007.html>)



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Operate the engine to 70°C [160°F] coolant temperature and check for leaks.



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